

UbiFix: Tackling Repairability Challenges in Smart Devices

Neelima Sailaja

Horizon Digital Economy Research,
University of Nottingham, United
Kingdom
neelima.sailaja@nottingham.ac.uk

Teresa Castle-Green

Horizon Digital Economy Research,
University of Nottingham, United
Kingdom
teresa.castle-
green@nottingham.ac.uk

Paul Coulton

Imagination Lancaster, Lancaster
University, Lancaster, United
Kingdom
p.coulton@lancaster.ac.uk

Michael Stead

Imagination Lancaster, University of
Lancaster, Lancaster, United Kingdom
m.stead@lancaster.ac.uk

Joseph Lindley

Imagination Lancaster, Lancaster
University, Lancaster, United
Kingdom
j.lindley@lancaster.ac.uk

Lachlan Urquhart

Department of Law,
Institution/University of Edinburgh,
United Kingdom
lachlan.urquhart@ed.ac.uk

Dimitrios Darzentas

School of Computing, Engineering
and the Built Environment (SCEBE),
Edinburgh Napier University, United
Kingdom
d.darzentas@napier.ac.uk

ABSTRACT

IoT products are increasingly becoming the default, with non-IoT versions of common hardware (e.g., TVs and printers) harder to find. Alongside this adoption surge, lack of support, outdated security, and planned obsolescence present concerning sustainability issues, contribute to eWaste growth and widen digital divides globally. This workshop aims to present and discuss legal, social, technical, and design aspects of repair practices, engaging the Ubicomp community in exploring challenges and opportunities for more repairable IoT devices. Focusing on diverse repair scenarios, the workshop seeks to establish a concise, holistic, and inclusive agenda for this research domain's future. Participants will map key research questions to support the movement towards more repairable technology.

CCS CONCEPTS

• **Human-centered computing:**

KEYWORDS

IoT, Sustainability, Repair, Smart Devices

ACM Reference Format:

Neelima Sailaja, Teresa Castle-Green, Paul Coulton, Michael Stead, Joseph Lindley, Lachlan Urquhart, and Dimitrios Darzentas. 2023. UbiFix: Tackling Repairability Challenges in Smart Devices. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing*

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-0200-6/23/10...\$15.00

<https://doi.org/10.1145/3594739.3605113>

& the 2023 ACM International Symposium on Wearable Computing (*UbiComp/ISWC '23 Adjunct*), October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3605113>

1 INTRODUCTION

As wireless connectivity and automation integrate into household items, IoT products increasingly become the norm. It's now harder to find non-IoT alternatives for devices like smart TVs and printers. Known IoT brands often offer lower prices to encourage adoption, such as Amazon Echo. Additionally, cheaper IoT brands can be found online, but their security and hardware quality may be questionable. While IoT adoption is currently a consumer choice, its rapid growth may lead to citizens having it imposed, like local governments monitoring homes [5]. Alongside the rise in IoT usage, sustainability concerns grow as well.

Repair and reuse of technology are essential response measures. Solutions involving repair span various societal levels, including legal actions like the Right to Repair or France's repairability scores on products, economic theories and methodologies within the Circular Economy, academic discussions, and the growing presence of repair cafes worldwide. IoT researchers and designers play a crucial role in these initiatives. They can support the Right to Repair through practical design recommendations for IoT devices (e.g., the Nokia G22 with a repair kit [6]); connect academic insights on the Circular Economy (and repair) to functional design solutions for current and future IoT devices; and assist community initiatives like Repair Cafes by providing design guidelines and guides for repairers and customers in their everyday repair experiences.

However, formal investigations into these possibilities have yet to be conducted. It is crucial to identify the key challenges, opportunities, and expectations of IoT researchers and designers regarding device repair, as it presents global technological, social, and environmental challenges. Ubicomp, with its diverse history of engaging

with embedded and ubiquitous technologies such as IoT and smart devices, serves as an ideal platform to initiate this effort. By gathering IoT enthusiasts with various skill sets and from different contexts, who share an interest in the repair and sustainability of smart devices, we can address these challenges more effectively.

The goal of this workshop is to start discussions about repair by examining current methods, obstacles, and support in IoT design, research, and practice. We will also explore future expectations and opportunities while relating it to the broader Ubicomp, HCI, and Sustainability zeitgeist. This will be a key step in moving the focus of IoT towards a more repair-focused, sustainable, and inclusive approach.

The workshop will be based on our diverse experience in repairing smart devices, covering legal, social, HCI, computing, and design aspects. This includes investigations around these aspects of this scenario, including a caravan experience that engages users in the act of repair at a community level [7].

We will share and compile various repair contexts for smart devices to create a comprehensive map of current processes and barriers. Our goal is to examine potential changes to the status quo, fostering sustainable design practices for future technologies. We will address specific questions during this workshop:

1. Should we prioritize repairability in designing IoT devices and smart technologies?
2. What are the main challenges in making IoT devices repairable by non-experts?
3. How can the Ubicomp community contribute to improving repairability?
4. What are the practical steps to implement these improvements?

The workshop comprises three stages. First, we will present the organizers' perspective on the current state of research in this field, allowing attendees to discuss the successes and failures in incorporating repairability into these contexts. Second, we will guide participants in examining practical design modifications and interventions that would effectively promote repair and support within sample technologies. Third, we will utilize the specific design processes, decisions, and revisions from the first two sections to generate a commentary on how repair has accelerated or impeded the design and adoption of Ubiquitous technologies. The goal is to create a succinct agenda for future research, identifying the questions that need to be addressed to advance the repair movement within technology.

Repair is crucial, but making it accessible and effective can be difficult. Additionally, moving from small-scale repair cafes to having a global influence demands persistent teamwork across different fields. Thus, the workshop's goal is to create a plan to enhance and encourage repair of Ubicomp technologies.

2 PRE-WORKSHOP PLANS

The workshop recruitment will take place via a website, offering two ways for participants to express interest: a traditional 'position paper' or a lighter online form on the website to share a repair-related provocation. The position paper must follow the UbiComp submission format, with a 2-page maximum, excluding references, to be submitted on PCS. The accepted papers will be included in the

ACM Digital Library and supplemental proceedings of the conference. The paper should depict a repair scenario involving a smart device, ranging from troubleshooting issues, lacking proper guidance or parts, successful or unsuccessful repair attempts, or even a fictional repair story illustrating a specific perspective on smart technology repair. Alternatively, participants may critique or comment on repair support and related practices for smart technologies.

The two-part approach aims to maximize inclusivity by accommodating those who prefer a formal contribution (through a workshop paper) and those who opt for a less academic path. This ensures a diverse range of repair contexts in the workshop, which is crucial for a comprehensive understanding and creating a representative, inclusive future agenda.

We will select participants based on their interest in repair processes and the quality of their submission related to the workshop theme. Our aim is to form a diverse group with varied perspectives, ensuring a dynamic and fruitful session. We will strive for inclusivity, considering factors such as career stage, race, gender, and location.

The participants would be provided access to a Slack channel before the workshop event for the dissemination of information and to provide a platform for the participants to introduce themselves and get involved with the community prior to the event itself, it will provide a persistent means to share content during the workshop and will be leveraged for future collaborations and report writing and special issue engagement post the workshop.

3 WORKSHOP STRUCTURE

The workshop is designed for a maximum of 30 participants and will consist of a single day 4-hour session, with two breaks. Limiting to 30 participants is necessary to ensure ample opportunity for all to participate.

3.1 Activity 1 : Orientation [30 minutes]

The workshop will commence with an introductory session outlining the challenges related to the sustainability of IoT and smart technologies. The organizing team will lead a presentation highlighting the workshop's objectives and themes. Following the presentation, a warm-up activity will be conducted to familiarize participants with one another and the workshop's goals. Additionally, this session will allocate time for obtaining participant consent before proceeding with the activities. The warm-up activity will involve an engaging ice-breaker, designed to encourage participation from all attendees. Utilizing a straightforward facilitation approach, participants will share their views in a concise manner by completing the phrases, "The best thing about being able to repair IoT is..." and "The worst thing about being able to repair IoT is...". This exercise will facilitate open discussion and set the tone for the rest of the workshop.

3.2 Activity 2 : State-of-the-art showcase [30 minutes]

In the second session, organizers will discuss repair as a solution to the issues addressed earlier, covering legal, HCI, and design perspectives. This will include an overview of the Right to Repair movement, its international manifestations, and specific country

Table 1: Summary of Workshop Activities

Activity	Time
Activity 1 : Orientation	30 minutes
Activity 2 : State-of-the-art showcase	30 minutes
Break	15 minutes
Activity 3 : Critique State-of-the-art	30 minutes
Activity 4 : Rebuild : Break and Make	30 minutes
Break	15 minutes
Activity 4 : Rebuild : Break and Make	15 minutes
Activity 5 : Feedback and Discussion	45 minutes
Activity 6 : Wrap up	minutes

**Figure 1: The IoT equipped caravan experience ©Hydar Dewachi**

implementations. Next, the session will explore user's engagement with sustainability and repair, examining its impact on IoT devices. The UK's Repair Cafe culture will be introduced, covering common principles, processes, and variants. Additionally, participants will be introduced to a unique caravan experience, which equips users with smart devices and encourages them to engage with repair and creation of simple technologies, presented via a live walkthrough or recorded video. Throughout this session, participants will receive materials to help them note their thoughts on the strengths and weaknesses of the various repair perspectives and examples presented. These insights will be utilized in the third activity.

3.3 Activity 3 : Critique State-of-the-art [30 minutes]

In this session, participants will engage in a group activity, with the larger group divided into smaller groups of 3-4 people to facilitate in-depth discussions and responses. Within their groups, participants will critique the state-of-the-art repair concepts presented earlier, grounding their arguments in the examples provided in their submissions. Each group will receive a starter kit containing a summary of the key points from the legal, design, and technical perspectives on repair. The kit will also include a guided worksheet for critiquing the concepts, offering space for participants

to record their notes and responses. This structured approach will help generate a well-rounded critique of the repair landscape.

3.4 Activity 4 : Rebuild : Break and Make [30 minutes]

Informed by the showcase and prior discussions, participants will now work in their groups to redesign a smart device, incorporating ideas generated from their critique in Activity 3. Each group will be provided with a popular IoT device, accompanied by a story outlining a repair need or fault. Participants will analyze the challenges presented in these repair scenarios and explore how they can apply their critiques to create better solutions. They will be encouraged to "break" conventional design norms and "make" new sustainable/inclusive approaches that enhance device repairability.

3.5 Feedback and Discussion [45 minutes]

In this session, each group will present their work to the entire room, covering the following aspects:

1. Their IoT device and repair scenario
2. The design revisions implemented for easier repair
3. Challenges encountered during these revisions
4. Exploring the balance between what needs to be done versus what can be done
5. An evaluation of how incorporating repair options improved the situation (or if not, why)

The wider group will have the opportunity to ask questions, raise comments, and discuss any concerns regarding the presented redesigns and repair scenarios. This interactive session will foster collaborative learning and generate further insights into IoT device repairability and design improvements.

3.6 Activity 6 : Wrap up Discussion [30 minutes]

In the final session, the entire group will reconvene to revisit the core research questions outlined in Section 1 of the workshop. The objective of this discussion is to delve deeper into these questions, refining and expanding upon them to gain a more detailed and nuanced understanding. These refined questions will serve as the foundation for post-workshop plans and establish the key elements of a new research agenda focused on the repair of IoT devices. This collaborative effort will ensure a well-rounded perspective

and a comprehensive approach to addressing the challenges and opportunities in IoT device repairability.

4 POST WORKSHOP PLANS

The workshop's goal is to spark dialogue surrounding repair in IoT devices, exploring how IoT enthusiasts can enable users to repair and reuse their devices rather than exclude and replace. We will use this as a foundation to propose an ACM Special Issue around Repair and Sustainability of SmartTech. The organizers' networks, along with the international UbiComp community's engagement, would be leveraged to successfully achieve this milestone, and form a foundational academic contribution towards the sustainability of IoT devices. We will publish the workshop outcomes as an academic paper and a report made publicly available on the HorizonDER [8] and Fixing the Future websites, and shared with project partners (e.g., BBC, Making Rooms Blackburn, Canadian Government, Which? etc.). This will advance research on repair in academia, support industry and independent initiatives (e.g., BBC's Sustainability initiatives, Restart Projects' FixFest[9], UK Repair Cafes). We will work with HorizonDER centre to produce a policy report to be submitted to future calls for evidence from the U.K. Parliament around Digital Exclusion.

5 ORGANISING COMMITTEE

Neelima Sailaja (Transitional Assistant Professor at the University of Nottingham) is an interdisciplinary researcher working on the socio-technical challenges of technology use. She currently leads the HDI wing of the EPSRC Fixing the Future project focusing on exploring the challenges and responses around repair of smart technologies. She has published over 15 peer reviewed outputs at leading HCI venues and has led workshops exploring the socio-technical implications of technology at ACM CHI and IMX.

Teresa Castle-Green is an inter-disciplinary researcher working on unpacking the socio-technical complexities of design and repair of IoT. She is the HDI Researcher at the University of Nottingham on the EPSRC Fixing the Future project engaging with UK repair communities to investigate ways in which HCI/HDI approaches can support the growing culture of community-based repair.

Paul Coulton is Professor of Speculative and Game Design at Lancaster University. Previously, he led Nokia's Academic Mobile Experiences Group (2006-2012). His current focus is the design challenges presented by IoT and datafication. He is PI on the EPSRC Experiencing the Future Mundane and Uncanny AI projects which explore AI data legibility, agency and negotiability.

Michael Stead is Lecturer in Sustainable Design Futures at Lancaster University. His interdisciplinary research interrogates the environmental and social impacts data-driven technologies pose for sustainability goals like Net Zero 2050 and the Circular Economy. Michael leads Lancaster's work package on the EPSRC Fixing the Future project. He has published over 25 peer reviewed Design/HCI outputs and lead-authored PETRAS' Little Book of Sustainability for the IoT.

Joseph Lindley is a specialist in Design Research and Design Fiction. His focus is on promoting these approaches as strategies to assist emerging technologies in delivering sustainable and equitable futures. He is a member of the ACM Future of Computing

Academy, an alumnus of the PETRAS Centre for Excellence for IoT Cybersecurity, and a UKRI Future Leaders Fellow.

Lachlan Urquhart is Senior Lecturer in Technology Law and HCI at the University of Edinburgh. He is Director of the Regulation and Design Lab, and Co-Director of SCRIPT and CRISP. He has published over 50 papers in computing, law and ethics exploring socio-technical aspects of designing, living with, and regulating emerging information technologies. He leads the EPSRC 'Fixing the Future: Right to Repair and Equal-IoT' project.

Dimitrios Darzentas is a lecturer at Edinburgh Napier University. His multidisciplinary work is situated at the intersection of HCI and Design (Mixed Reality Technologies, Experience Design, Physical/Digital Service Design, Sustainability). His current research interests include Hybrid Physical/Digital Experiences and Socio-Political and Cultural Heritage aspects of Gaming.

ACKNOWLEDGMENTS

This research is supported by funding from the UKRI EPSRC [EP/W024780/1, EP/T022493/1].

REFERENCES

- [1] Alexander J.A.M. van Deursen, Alex van der Zeeuw, Pia de Boer, Giedo Jansen, and Thomas van Rompay. 2019. Digital inequalities in the Internet of Things: differences in attitudes, material access, skills, and usage. <https://doi.org/10.1080/1369118X.2019.1646777> 24, 2: 258–276. <https://doi.org/10.1080/1369118X.2019.1646777>
- [2] Vanessa Forti, Cornelius P Balde, Ruediger Kuehr, and Garam Bel. 2020. The Global E-waste Monitor 2020: Quantities, flows and the circular economy potential. Retrieved March 30, 2023 from <https://collections.unu.edu/view/UNU:7737>
- [3] Mohammad Rashidujjaman Rifat, Hasan Mahmud Prottoy, and Syed Ishtiaque Ahmed. 2019. The breaking hand: Skills, care, and sufferings of the hands of an electronic waste worker in Bangladesh. Conference on Human Factors in Computing Systems - Proceedings. <https://doi.org/10.1145/3290605.3300253>
- [4] Jinglei Yu, Eric Williams, Meiting Ju, and Yan Yang. 2010. Forecasting global generation of obsolete personal computers. Environmental Science and Technology 44, 9: 3232–3237. <https://doi.org/10.1021/es903350q>
- [5] UK councils deploy IoT in housing to boost safety and wellbeing - ITU Hub. Retrieved March 30, 2023 from <https://www.itu.int/hub/2021/01/uk-councils-deploy-iot-in-housing-to-boost-safety-and-wellbeing/>
- [6] Making self-repair simpler. Retrieved March 30, 2023 from https://www.nokia.com/phones/en_gb/self-repair
- [7] The Repair Shop 2049. Retrieved March 30, 2023 from <http://imagination.lancaster.ac.uk/project/the-repair-shop-2049/>
- [8] Horizon. Retrieved March 30, 2023 from <https://www.horizon.ac.uk/>
- [9] Fixfest. Retrieved March 30, 2023 from <https://fixfest.therestartproject.org/>

APPENDIX

A APPENDIX A

A.1 Call for Participation

We welcome you to UbiFix 2023, the first international workshop that explores the legal, technical, social and design aspects of IoT repair and sustainability.

IoT products are increasingly becoming the default, with non-IoT versions of common hardware (e.g., TVs and printers) harder to find. Alongside this adoption surge, lack of support, outdated security, and planned obsolescence present concerning sustainability issues, contribute to eWaste growth and widen digital divides globally.

This workshop aims to present and discuss legal, social, technical, and design aspects of repair practices, engaging the UbiComp community in exploring challenges and opportunities for more repairable IoT devices. The workshop will orient you to the current

literature and practice around repair of IoT devices; welcome your critique on the state-of-art; present examples and contexts of IoT repair and sustainability to support the critique; and encourage you to imagine a future where current challenges are barriers are overcome to enable better repairability of IoT.

For participation, you can either upload a two-page position paper (excluding references; UbiComp Submission Template) on the PCS website or submit a lighter online form to share a repair-related provocation. . The accepted papers will be included in the ACM Digital Library and supplemental proceedings of the conference. The paper could depict a repair scenario involving a smart device, ranging from troubleshooting issues, lacking proper guidance or parts, successful or unsuccessful repair attempts, or even a fictional repair story illustrating a specific perspective on smart technology repair. Alternatively, you may critique or comment on repair support and related practices for smart technologies.

A diverse group of complementary participants will be selected based on their prior experience of IoT repair and sustainability (or related areas) and the quality/ relevance of their submission to the workshop theme.

Upon acceptance, it is mandatory that at least one author of each accepted position paper must attend the workshop. More details are available at <https://ubifix2023.wp.horizon.ac.uk>.

A.1.1 Best Paper Award. A Best Paper Award will be given to the best submission made to UbiFix 2023. The awardee will be presented with a certificate and small memento at the workshop.

A.1.2 Key Dates [23:59 AoE for all Deadlines]. Paper Submission: 30th June 2023

Notification: 15th July 2023

Camera-Ready Version: 31st July 2023

Workshop: 8th/9th [TBD] October 2023

A.1.3 Workshop Organisers. Neelima Sailaja, University of Nottingham

Teresa Castle-Green, University of Nottingham

Joseph Lindley, Lancaster University

Micheal Stead, Lancaster University

Paul Coulton, Lancaster University

Lachlan Urquhart, University of Edinburgh

Dimitrios Darzentas, Napier University

A.1.4 Extended list of partners. BBC R&D

NCC Group

Which?

Active Ingredient

Federal Government of Canada

The Making Rooms

A.1.5 Contact Details: For any questions, queries or comments please email or .

Alternatively, you could also visit our workshop website for more information at <https://ubifix2023.wp.horizon.ac.uk>